

Final Project: Data Analysis & Visualization Dashboard

Course: Data Analysis and Visualization

You may choose one of the following domains:

- **Business Analytics**
- **Healthcare Analytics**

Programming-Based

- Python
- Jupyter Notebook
- Streamlit
- Dash

Visualization Platforms

- Power BI
- Tableau
- Looker Studio
- Dataset Requirements

Each group must use a dataset containing:

- Minimum **5,000 rows**
- Minimum **10 features/columns**
- Real-world data preferred
- CSV or Excel format

Project Tracks

Option 1: Business Dashboard

Suggested Business Topics

You may choose one:

- Sales Analytics
- E-commerce Analytics
- Customer Behavior Analysis
- Financial Dashboard
- Marketing Analytics
- Retail Analytics
- Supply Chain Dashboard

Required Dashboard Components

1. KPI Section

Include:

- Total Sales
- Total Profit
- Average Revenue
- Customer Count
- Monthly Growth
- Best-Selling Product

2. Required Visualizations

You must include at least:

- Bar Chart
- Line Chart
- Pie/Donut Chart
- Heatmap
- Scatter Plot
- Geographic Map (optional)
- Time-Series Visualization

3. Interactive Features

Dashboard must contain:

- Filters
- Dropdown menus

- Search functionality
- Dynamic charts
- Date selection

4. Business Insights

You must answer questions such as:

- Which products generate the highest revenue?
- Which months have the highest sales?
- Which regions perform best?
- What customer patterns exist?
- Which categories are underperforming?

Option 2: Healthcare Dashboard

Suggested Healthcare Topics

You may choose one:

- Hospital Management
- Disease Analytics
- Patient Monitoring
- Cancer Analysis
- Diabetes Dashboard
- COVID-19 Analysis
- Public Health Analytics
- Healthcare Resource Management

Required Dashboard Components

1. KPI Section

Include:

- Total Patients
- Disease Count
- Mortality/Recovery Rate
- Admission Rate

- Gender Distribution
- Age Group Analysis

2. Required Visualizations

You must include:

- Disease Distribution Chart
- Age Distribution Histogram
- Correlation Heatmap
- Trend Analysis
- Patient Outcome Visualization
- Geographic Distribution (if available)

3. Interactive Features

Dashboard should include:

- Patient filtering
- Disease selection
- Time filtering
- Dynamic statistics
- Drill-down analysis

4. Healthcare Insights

You must answer questions such as:

- Which disease is most common?
- Which age groups are most affected?
- What trends exist over time?
- Are there seasonal patterns?
- Which risk factors are important?

Technical Requirements

Data Processing

You must:

- Handle missing values
- Remove duplicates
- Detect outliers
- Normalize/transform data if needed

Exploratory Data Analysis (EDA)

You must perform:

- Statistical summaries
- Distribution analysis
- Correlation analysis
- Trend analysis

Dashboard Design Requirements

The dashboard must:

- Be visually appealing
- Use consistent colors/themes
- Include titles and labels
- Be easy to navigate
- Support interactive exploration

Deliverables

You must submit:

1. Final Report (PDF/Word)

The report should include:

- Introduction
- Dataset description
- Methodology

- Data preprocessing
- Visualization techniques
- Dashboard screenshots
- Insights and findings
- Conclusion

2. Source Code

Submit:

- Python notebooks/scripts
- Dashboard files
- Power BI/Tableau files

Bonus Marks (Optional)

You can earn bonus points for:

- Machine Learning Integration
- Predictive Analytics
- AI Chatbot Integration
- Real-Time Dashboard
- Deployment to Web
- Mobile-Friendly Dashboard

Suggested Public Datasets

Business Datasets

- [Kaggle Business Datasets](#)
- [Google Dataset Search](#)

Healthcare Datasets

- [UCI Machine Learning Repository](#)

- [WHO Open Data](#)
- [CDC Open Data](#)